

Topic 2



Lecture 2 · Exploratory Data Analysis (EDA) Part 1

National University of Singapore (NUS)

TODAY'S OBJECTIVES

1

What is a Variable

2

Types of Variables

- Quantitative: Discrete vs Continuous
- Categorical: Nominal vs Ordinal

3

Single Variable Exploration

- Frequency Table
- Bar Plot
- Histogram

01

What is a Variable?

Variable



DEFINITION

A variable is any characteristic observed in a study.



KEY INSIGHT

Values of variables can vary from one observation to another.

EXAMPLE

Study: Smoking & Lung Cancer

Variables:

- Number of cigarettes smoked in a day
- Gender (male / female)
- Age (in years)
- Lung cancer status (yes / no)

02

Types of Variables

Quantitative vs Categorical Variables

Categorical vs Quantitative Variables

Categorical

Each observation belongs to one of a set of categories.

Numbers used as labels only (e.g. 0 = male, 1 = female).

Examples: Gender, Race, Smoking status, Lung cancer status

Quantitative

Observations take numerical values

Numerical values representing a magnitude.

Examples: Age, Height, Weight, Blood pressure, Income

How to Tell Them Apart

Ask: Is the difference between two values meaningful?

Categorical: Gender

Male: 0

Female: 1

$$1 - 0 = 1$$

no real-world interpretation



KEY INSIGHT

The presence of numbers doesn't make a variable quantitative. The meaning of those numbers does.

Quantitative: Blood Pressure

Person A: 100

Person B: 120

$$120 - 100 = 20$$

clear interpretation: Person B has a blood pressure higher than Person A by 20 units

Quantitative Variables: Discrete vs. Continuous

Discrete

Possible values are separated by gaps.

Examples:

- No. of pets at home
- No. of children
- No. of emails received



KEY INSIGHT

Often but **not always whole numbers**: even decimals can be discrete.

Example: rod lengths of 1.0, 1.5, 2.0cm (gaps of 0.5)

The key is gaps exist.

Continuous

Values can exist anywhere within a continuous range (infinitely many possible values).

Examples:

- Height
- Weight
- Blood pressure

Continuous variables may *appear discrete* due to limitations in measuring instruments

Weight is continuous by nature

Continuous range of values: 65.123456
65.123457



Weight appears discrete



Records to 1 decimal place

Possible Values: 65.0
65.1

In practice, statisticians classify a variable as continuous if:

- Measurement scale is sufficiently fine
- Classification will not meaningfully affect results

In this course, we will focus on variables that are continuous by nature

Categorical Variables: Nominal vs. Ordinal

Nominal

Categories have no natural ordering.

Labels are simply different groups: no ranking implied.

Examples:

- Gender (male / female)
- Race (Chinese / Malay / Indian / Others)
- Pregnancy status (yes / no)

There is no 'greater' or 'lesser' category.

Ordinal

Categories have a meaningful natural order.

Differences between categories are NOT measured numerically.

Examples:

- Ratings: bad < normal < good
- Competition: 1st > 2nd > 3rd
- Age group: 25–45 < 46–65 < 65+

Order exists, but exact gaps between categories are undefined.

Representing the Same Variable in Different Types

1

Variable: Age

Quantitative
Continuous
Variable



2

**Group into
Age Ranges:**

- 25 – 45
- 46 – 65
- Above 65



3

Categorise

Count how many
participants fall into
each age group

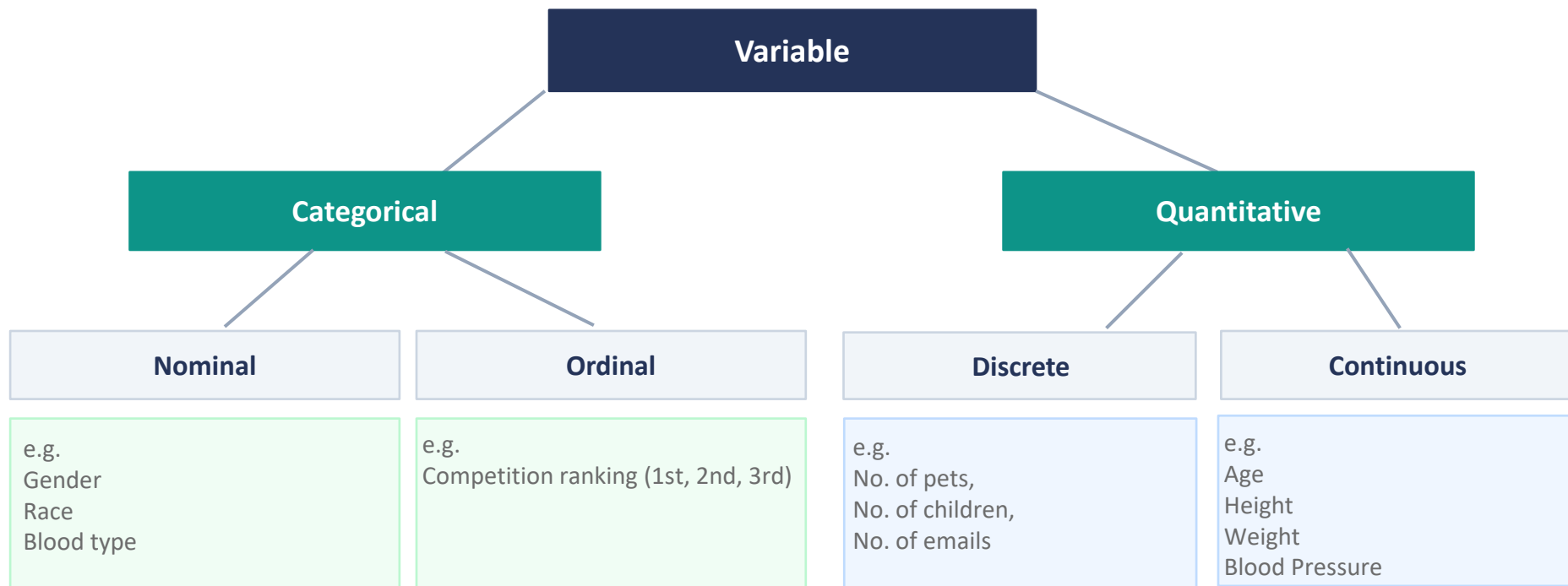


4

Variable: Age

Categorical
Ordinal
Variable

Types of Variables: Overview



03

Single Variable Exploration

Frequency Table, Bar Plot, Histogram

Frequency Table



DEFINITION

A table that lists all the possible values or categories of a variable together with the number of observations in each category.

KEY TERMS

Frequency: count of observations

Proportion: $\text{count} \div \text{total}$

Percentage: $\text{proportion} \times 100$

Relative Frequency: percentage or proportion

Modal category: category with the highest frequency

Example 1: Race

Variable **Race** in a sample of 100 Singaporeans:

Race	Count	Proportion	Percentage
Chinese	76	0.76	76%
Malay	15	0.15	15%
Indian	8	0.08	8%
Others	1	0.01	1%
Total	100	1.00	100%

Modal Category

Chinese

Highest frequency: 76 out of 100

Example 2: Lung Cancer Data

Study

Investigate the effect of smoking on the incidence of lung cancer

Sample

17 lung cancer patients

17 healthy people

Variable	Description	Values	Type
Age	Age at time of study (years)	Numerical	Quantitative Continuous
Smoke	Smoking status	0 = No, 1 = Yes	Categorical Nominal
Cancer	Lung cancer status	0 = No, 1 = Yes	Categorical Nominal
Gender	Gender of participant	0 = Female, 1 = Male	Categorical Nominal

Example 2: Lung Cancer Data — Gender

Count Table

Gender	Count
Male	18
Female	16

Proportion Table

Gender	Proportion
Male	0.529412
Female	0.470588

Percentage Table

Gender	Percentage
Male	52.9412
Female	47.0588

Example 2: Lung Cancer Data — Cancer

Count Table

Cancer	Count
No	17
Yes	17

Proportion Table

Cancer	Proportion
No	0.50
Yes	0.50

Percentage Table

Cancer	Percentage
No	50.0
Yes	50.0

Example 2: Lung Cancer Data — Age

Count Table

Age	Count
25 - 45	10
46 - 65	15
more than 65	9

Proportion Table

Age	Proportion
25 - 45	0.294118
46 - 65	0.441176
more than 65	0.264706

Percentage Table

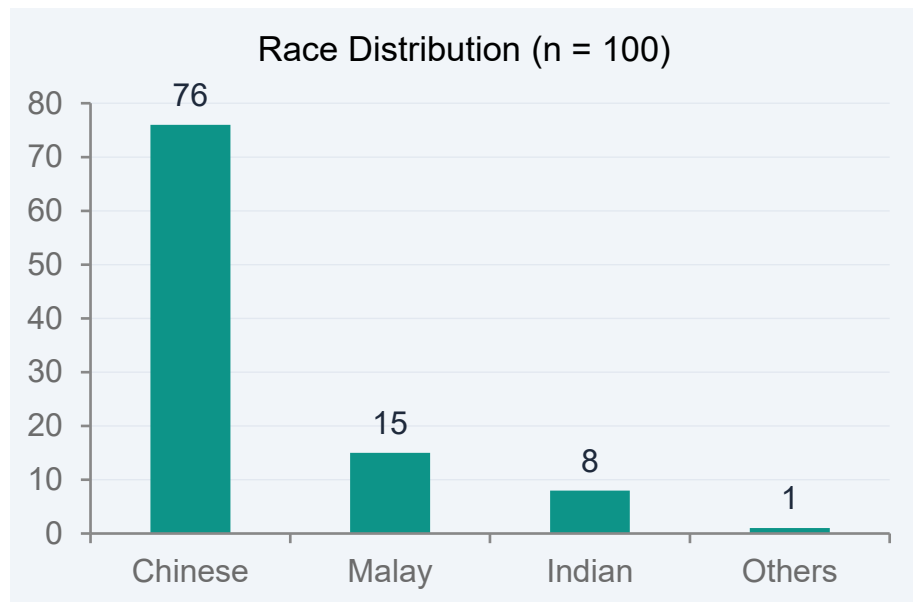
Age	Percentage
25 - 45	29.4118
46 - 65	44.1176
more than 65	26.4706

Summarising a Frequency Table

Mention:

- 1 Modal category
- 2 The proportion or percentage for modal category

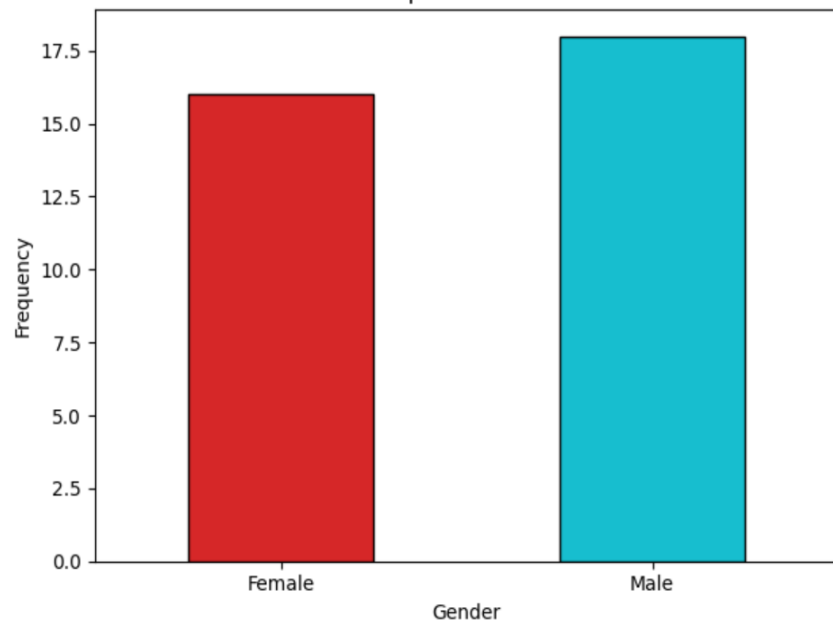
Bar Plot



- **Common way to display a single categorical variable**
- **Vertical Bar for each category**
- **Height = frequency, proportion, or percentage**
The y-axis can represent any of these.
- **Modal category = tallest bar**
Here: Chinese (76 out of 100).

Example 1: Lung Cancer Data — Gender

Bar plot of Gender

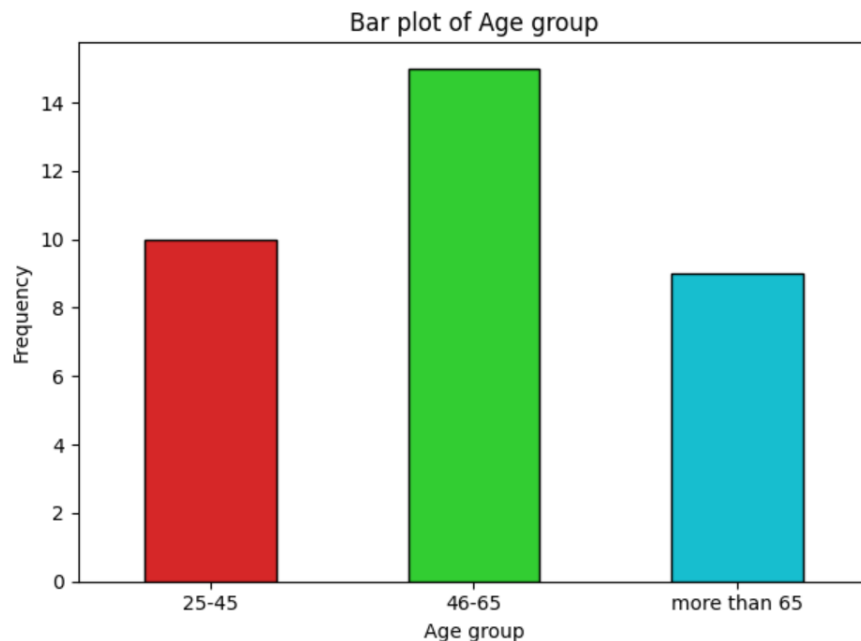


Modal Category

Male

Highest frequency: 18

Example 1: Lung Cancer Data — Age



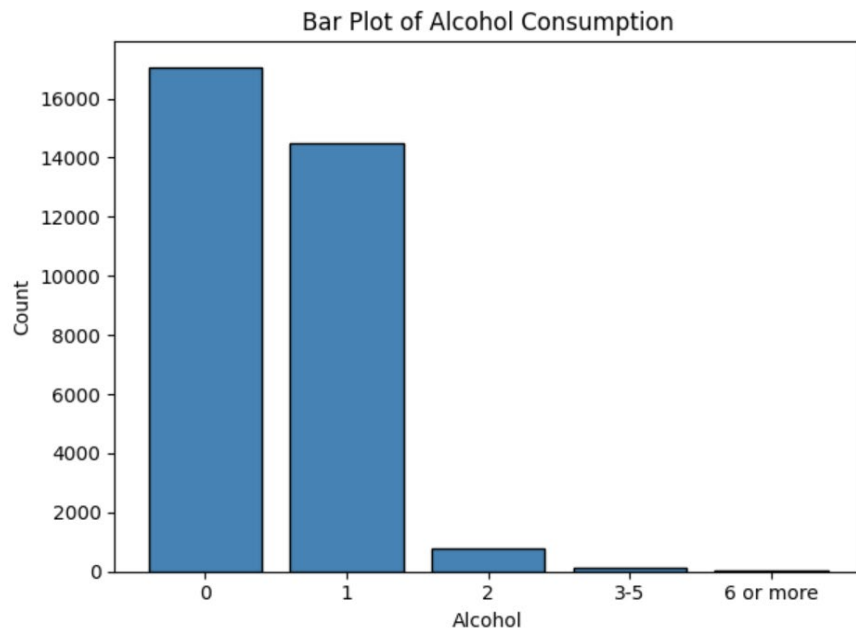
Modal Category

45 - 65

Highest frequency: 15

Example 2: Alcohol Consumption & Pregnancy

Variable: Mother's level of alcohol consumption during pregnancy



Modal Category

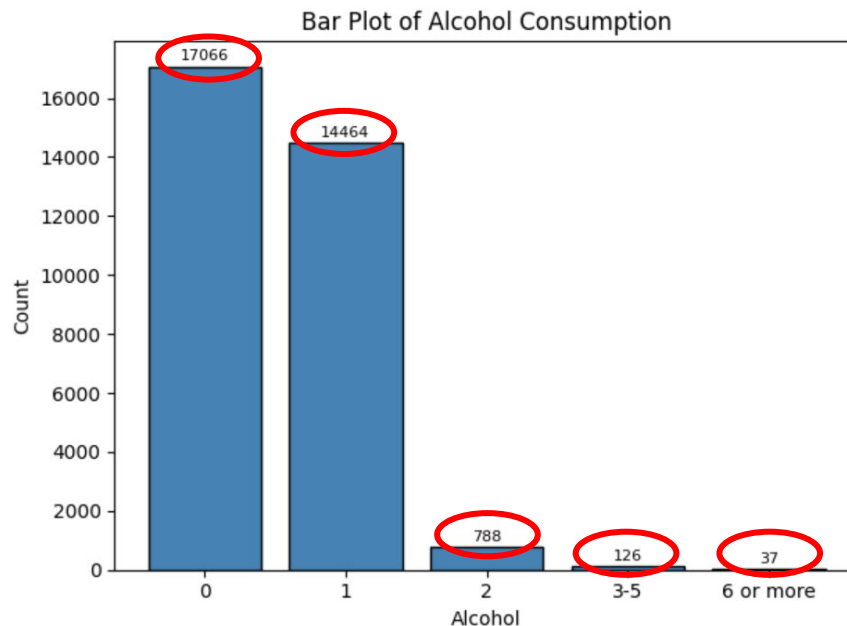
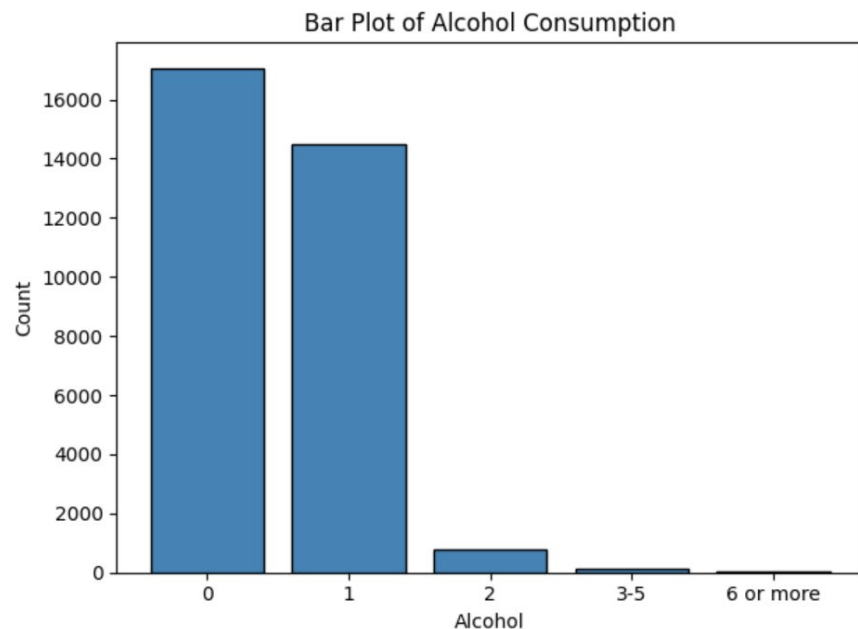
0

Overall Trend

As the level of alcohol consumption increases, the number of mothers in each category decreases

Example 2: Alcohol Consumption & Pregnancy

Use of labels to improve readability of graph



Summarising a Bar Plot

Mention:

- 1 Modal category, together with its proportion or percentage
- 2 Any groups of categories with high or low proportions
- 3 If the categories have a natural ordering, should comment on whether the frequencies follow any noticeable pattern.

Histogram



DEFINITION

A graphical display of a quantitative variable using bars, where the height of each bar represents the frequency or relative frequency of observations in each interval.

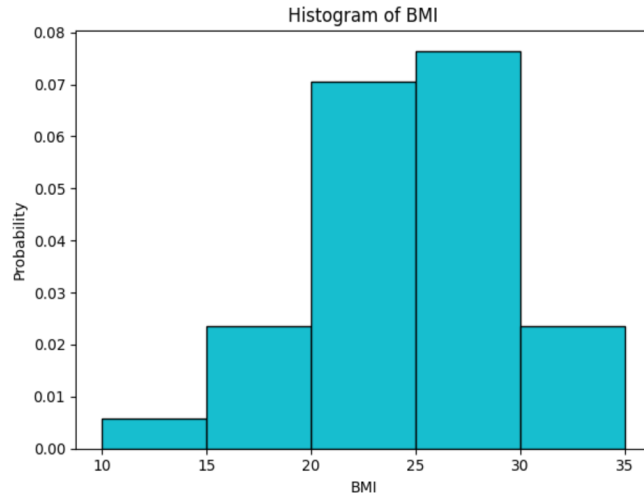
Creating a Histogram

STEP-BY-STEP

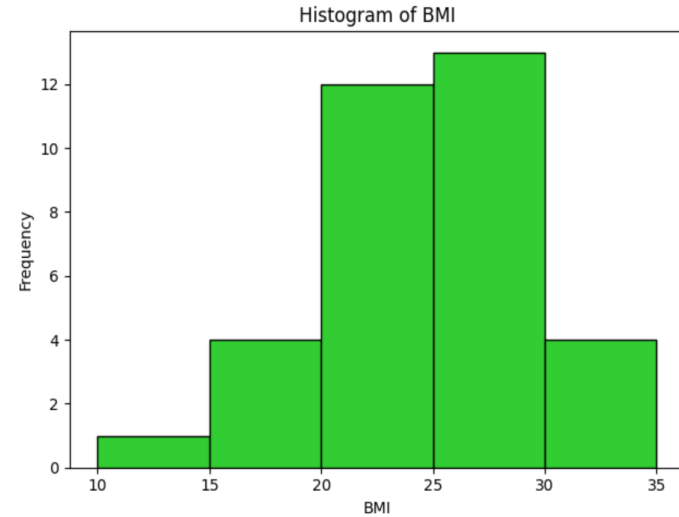
- 1 Divide the range of the data into intervals of equal width
- 2 Count how many observations fall within each interval.
- 3 label the intervals on the x-axis.
- 4 Draw bars whose heights represent the frequencies or relative frequencies.

Histogram

Probability



Frequency



- 1 person has a BMI 10 - 15
- 4 people have BMI 15 - 20

Summarising a Histogram

Mention:

1

Overall pattern of the distribution:

Do most observations cluster together, or are there large gaps between observations? Any Outliers?

2

Number of peaks or “mounds”:

Unimodal, Bimodal or Multimodal?

3

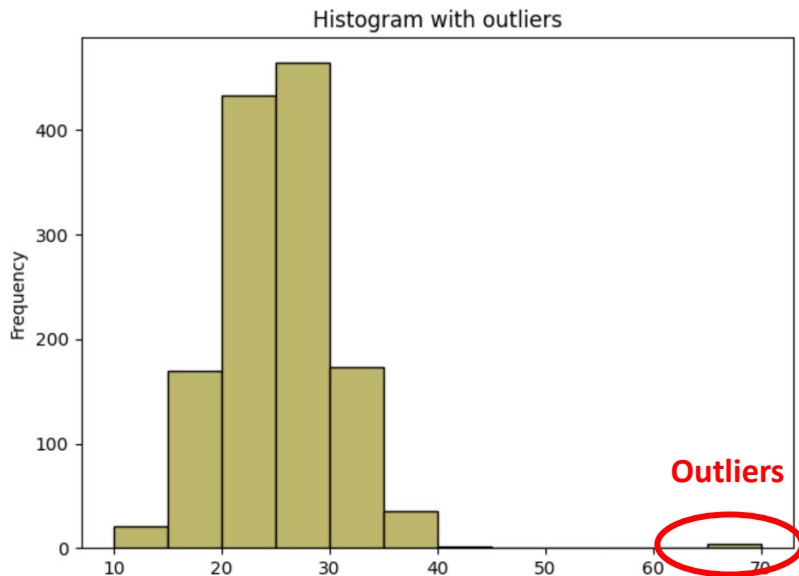
Skewness:

Is the data distribution symmetric or skewed?

1

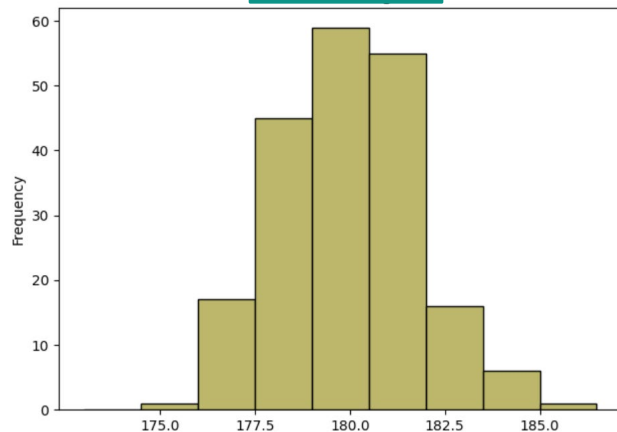
Overall pattern of the distribution:

Do most observations cluster together, or are there large gaps between observations? Any Outliers?

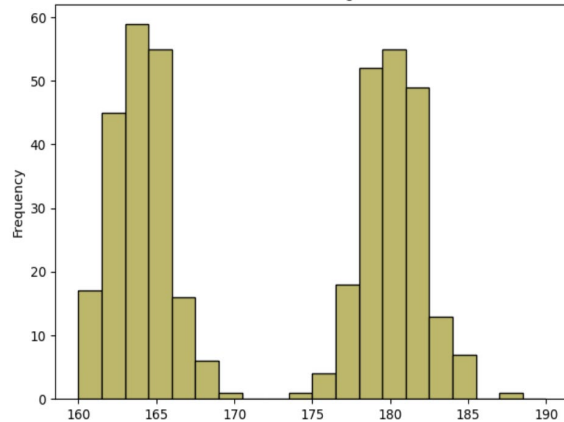
**KEY INSIGHT**

A large gap followed by a few isolated bars may suggest the presence of suspected outliers.

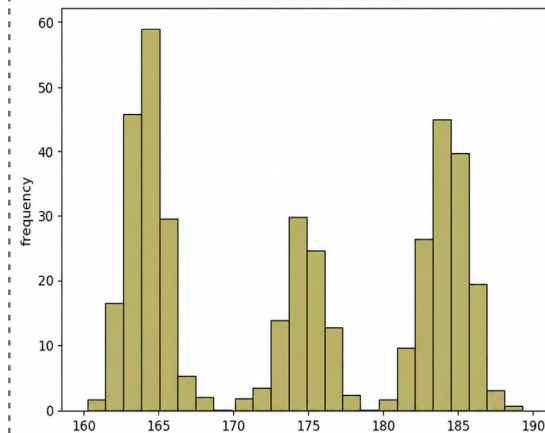
2 Number of peaks or “mounds”: Unimodal, Bimodal or Multimodal?

Unimodal

One peak / mound

Bimodal

Two peaks / mounds

Multimodal

More than two peaks

3

Skewness:

Is the data distribution symmetric or skewed?

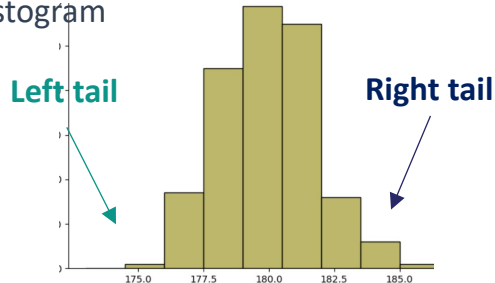
**DEFINITION**

A distribution is **skewed to the left** if the left tail is longer than the right tail

A distribution is **skewed to the right** if the right tail is longer than the left tail

KEY TERMS

Tail: Ending part of a histogram

**NOTE:**

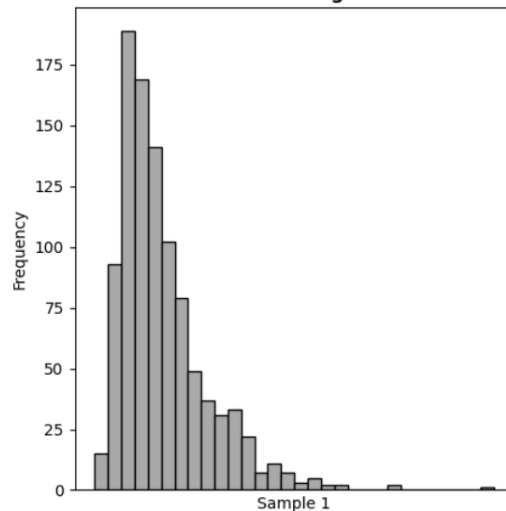
When describing the tail of a histogram or a distribution, some terminologies are used interchangeably:

- longer = thicker = heavier
- shorter = thinner = lighter

3

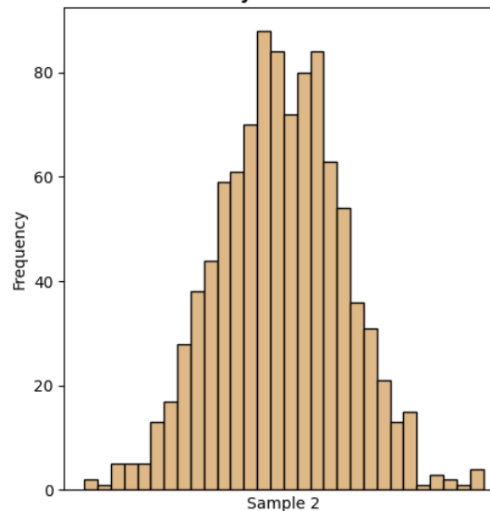
Skewness:

Is the data distribution symmetric or skewed?

Skewed Right

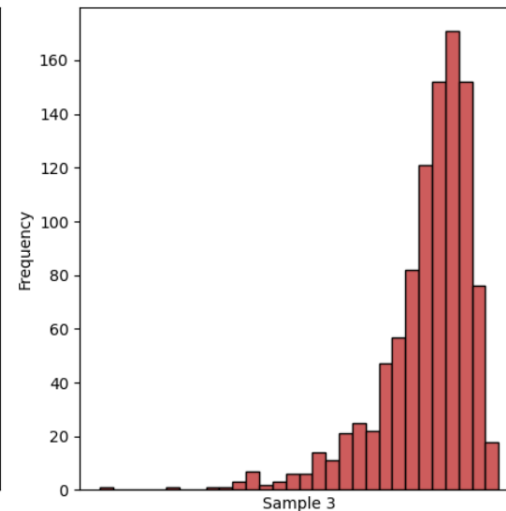
Right tail is longer than left

Example: Income

Symmetric

Both tails roughly balanced

Example: IQ

Skewed Left

Left tail is longer than right

Example: Lifespan